

SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:
US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

Issuing Date 22-Oct-2025

Revision Date 22-Oct-2025

Revision Number 2

1. Identification

Product identifier

Product Name PROTHIOGUARD FUNGICIDE

Other means of identification

Product Code 2123310

Synonyms None

Recommended use of the chemical and restrictions on use

Recommended Use Agriculture

Restriction on Use Do not use where contact with food or drinking water is possible Avoid prolonged contact with eyes, skin, and clothing

Details of the supplier of the safety data sheet

Supplier Address

ACETO LIFE SCIENCES, L.L.C.
4 Tri Harbor Court
Port Washington, NY 11050-4661
Phone: (516) 627-6000
Fax: (516) 627-6093
Email: aceto@aceto.com

Emergency telephone number

Emergency telephone number(s) ChemTrec: 1-800-424-9300
Outside of the United States: 1-703-527-3887

2. Hazard(s) identification

Classification

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Hazards not otherwise classified (HNOC)

Not applicable

Label elements

Hazard statements

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Other Information

No information available.

3. Composition/information on ingredients

Substance

Synonyms Not available

Formula Not available

Chemical name	CAS No.	Weight-%	Trade secret
Prothioconazole Technical	178928-70-6	41	

4. First-aid measures

Description of first aid measures

Inhalation	Remove to fresh air.
Eye contact	Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.
Skin contact	Wash skin with soap and water.
Ingestion	Rinse mouth.

Most important symptoms and effects, both acute and delayed

Symptoms	No information available.
Effects of Exposure	No information available.

Indication of any immediate medical attention and special treatment needed

Note to physicians	Treat symptomatically.
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5. Fire-fighting measures

Suitable Extinguishing Media	Alcohol resistant foam. Carbon dioxide (CO ₂). Dry chemical. Water spray.
Unsuitable extinguishing media	No information available.
Specific hazards arising from the chemical	No information available.
Hazardous Combustion Products	Carbon oxides. Nitrogen oxides (NO _x). Sulfur Dioxide. Hydrogen Cyanide. Hydrogen Chloride.
Explosion Data	
Sensitivity to mechanical impact	None.
Sensitivity to static discharge	None.
Special protective equipment and precautions for fire-fighters	Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

Personal precautions Ensure adequate ventilation.

Methods and material for containment and cleaning up

Methods for containment Prevent further leakage or spillage if safe to do so.

Methods for cleaning up Pick up and transfer to properly labeled containers.

7. Handling and storage

Precautions for safe handling

Advice on safe handling Handle in accordance with good industrial hygiene and safety practice.

Conditions for safe storage, including any incompatibilities

Storage Conditions Keep container tightly closed in a dry and well-ventilated place.

8. Exposure controls/personal protection**Control parameters**

Exposure Limits This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

Chemical name	ACGIH TLV	OSHA PEL	NIOSH
Prothioconazole Technical 178928-70-6	-	-	-

Appropriate engineering controls

Engineering controls Showers
Eyewash stations
Ventilation systems.

Individual protection measures, such as personal protective equipment

Eye/face protection No special protective equipment required.

Gloves

Skin and body protection No special protective equipment required.

Respiratory protection No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

General hygiene considerations Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Information on basic physical and chemical properties

Physical state	Liquid
Appearance	Opaque White liquid
Color	Opaque; White
Odor	Mildly Bitter
Odor Threshold	No information available

Property	Values	Remarks • Method
pH	5.88 (1% dispersion)	None known
pH (as aqueous solution)		None known
Melting point / Freezing point	No data available	None known
Boiling point / boiling range	No data available	None known
Flash Point	> 110 °C / >230 °F	None known
Evaporation Rate	No data available	None known
Flammability (solid, gas)	No data available	None known
Flammability Limit in Air		None known
Upper flammability limit	No data available	
Lower flammability limit	No data available	
Vapor Pressure	No data available	None known
Vapor Density	No data available	None known
Relative density	No data available	None known
Water solubility	No data available	Dispersible
Solubility in other solvents	No data available	None known
Partition coefficient: n-octanol/water	No data available	None known
Autoignition temperature °C	No data available	No information available
Decomposition temperature		None known
Kinematic Viscosity	No data available	None known
Dynamic Viscosity	No data available	None known

Other information

Explosive properties	No information available
Oxidizing properties	No information available
Softening Point	No information available
Molecular Weight	No information available
VOC content	No information available
Liquid Density	1.13 g/cm ³ (9.43 pounds/gallon) at 25 C
Bulk density	No information available

10. Stability and reactivity

Reactivity	No information available.
Chemical stability	Stable under normal conditions.
Possibility of hazardous reactions	None under normal processing.
Conditions to Avoid	Extremes of temperature and direct sunlight. Do not freeze.
Incompatible materials	None known based on information supplied.
Hazardous decomposition products	Carbon oxides. Nitrogen oxides (NOx). Sulfur Dioxide. Hydrogen Chloride. Hydrogen cyanide (hydrocyanic acid).

11. Toxicological information

Information on likely routes of exposure

Inhalation	Harmful if inhaled.
Eye contact	Moderately irritating to the eyes.
Skin contact	Substance may cause slight skin irritation.
Ingestion	Harmful if swallowed.

Symptoms related to the physical, chemical and toxicological characteristics

Symptoms No information available.

Acute toxicity .

Numerical measures of toxicity

No information available

LD50 (female Rat) > 2,000 - < 5,000 mg/kg LD50 (male/female combined Rat) > 5,000 mg/kg LC50 (male/female combined Rat) > 1.6 mg/l Exposure time: 4 h Determined in the form of liquid aerosol. Highest attainable concentration. No deaths

Chemical name	Oral LD50	Dermal LD50	Inhalation LC50
Prothioconazole Technical 178928-70-6	-	-	-

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Skin corrosion/irritation No information available.

Serious eye damage/eye irritation No information available.

Respiratory or skin sensitization No information available.

Germ cell mutagenicity No information available.

Carcinogenicity No information available.

Chemical name	ACGIH	IARC	NTP	OSHA
Prothioconazole Technical 178928-70-6	-	-	-	-

Reproductive toxicity No information available.

STOT - single exposure No information available.

STOT - repeated exposure No information available.

Aspiration hazard No information available.

Other Adverse Effects No information available.

Interactive effects No information available.

12. Ecological information

Ecotoxicity

Chemical name	Algae/aquatic plants	Fish	Toxicity to microorganisms	Crustacea
Prothioconazole Technical 178928-70-6	IC50 (Raphidocelis subcapitata (freshwater green alga)) 2.18 mg/l Growth rate; Exposure time: 72 h The value mentioned relates to the active ingredient prothioconazole. ErC50 (Skeletonema costatum) 0.03278 mg/l Exposure time: 72 h The value mentioned relates to the active ingredient prothioconazole. EC10 (Skeletonema costatum) 0.01427 mg/l Growth rate; Exposure time: 72 h The value mentioned relates to the active ingredient prothioconazole.	LC50 - Oncorhynchus mykiss (rainbow trout) - 1.83 mg/l - 96.0 h	-	EC50 - Daphnia magna (Water flea) - 1.2 mg/l - 48 h

Persistence/Degradability No information available.

Bioaccumulation There is no data for this product.

Chemical name	Partition coefficient
Prothioconazole Technical 178928-70-6	-

Other adverse effects No information available.

13. Disposal considerations

Waste treatment methods

Waste from residues/unused products Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging Do not reuse empty containers.

14. Transport information

DOT Not regulated

TDG

UN-No UN3082
 Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s.
 Transport hazard class(es) 9
 Packing Group III
 Special Provisions 16, 99
 Description UN3082, Environmentally hazardous substance, liquid, n.o.s. (Prothioconazole Technical), 9, III

MEX

UN-No UN3082
 Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s.
 Transport hazard class(es) 9
 Packing Group III
 Technical Name Prothioconazole Technical
 Description UN3082, Environmentally hazardous substance, liquid, n.o.s. (Prothioconazole Technical), 9, III
 Special Provisions 274, 331, 335

ICAO

UN-No UN3082
 Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s.
 Transport hazard class(es) 9
 Packing Group III
 Description UN3082, Environmentally hazardous substance, liquid, n.o.s. (Prothioconazole Technical), 9, III
 Special Provisions A97, A158, A197, A215

IATA

UN number or ID number UN3082
 Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s.
 Transport hazard class(es) 9
 Packing group III
 Technical Name Prothioconazole Technical
 Description UN3082, Environmentally hazardous substance, liquid, n.o.s. (Prothioconazole Technical), 9, III
 Special Provisions A97, A158, A197
 ERG Code 9L

IMDG

UN number or ID number UN3082
 Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s.
 Transport hazard class(es) 9
 Packing Group III
 EmS No. F-A, S-F
 Special Provisions 274, 335, 969
 Marine pollutant NP
 Description UN3082, Environmentally hazardous substance, liquid, n.o.s. (Prothioconazole Technical), 9, III, Residue: Last Contained

15. Regulatory information**International Inventories**

TSCA

Complies

Chemical name	CAS No.	Inventory Listing Status	Commercial Activity Designation
Prothioconazole Technical	178928-70-6	-	Unknown *

Chemical name	CAS No.	References
Prothioconazole Technical	178928-70-6	-

DSL/NDL	Does not comply
EINECS/ELINCS	Does not comply
ENCS	Does not comply
IECSC	Does not comply
KECL	Does not comply
PICCS	Does not comply
AIIC	Does not comply
NZIoC	Complies

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

EINECS/ELINCS - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

ENCS - Japan Existing and New Chemical Substances

IECSC - China Inventory of Existing Chemical Substances

KECL - Korean Existing Chemicals Inventory

PICCS - Philippines Inventory of Chemicals and Chemical Substances

AIIC - Australian Inventory of Industrial Chemicals

NZIoC - New Zealand Inventory of Chemicals

Federal Regulations

SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

CWA (Clean Water Act)

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

Chemical name	CWA - Reportable Quantities	CWA - Toxic Pollutants	CWA - Priority Pollutants	CWA - Hazardous Substances
Prothioconazole Technical 178928-70-6	-	-	-	-

CERCLA

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

Chemical name	Hazardous Substances RQs	Extremely Hazardous Substances RQs	Reportable Quantity (RQ)
Prothioconazole Technical 178928-70-6	-	-	

US State Regulations

California Proposition 65

This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know Regulations

This product does not contain any substances regulated under applicable state right-to-know regulations

Chemical name	New Jersey	Massachusetts	Pennsylvania
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Prothioconazole Technical 178928-70-6	-	-	-
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U.S. EPA Label Information**EPA Pesticide Registration Number** 2749-652**EPA Statement**

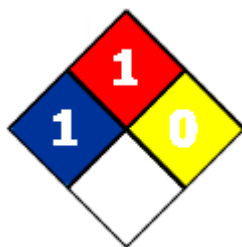
In the United States this product is regulated by the US Environmental Protection Agency (EPA) under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA). It is a violation of Federal law to use this product in a manner inconsistent with its labeling. read and follow all label directions. This product is excluded from listing requirements under EPA/TSCA. This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required by Safety Data Sheets, and for workplace labels of non-pesticide chemicals.

Other Regulations

US State Right-To-Know Ingredients : Glycerine (56-81-5) MN , RI.

16. Other information

NFPA **Health hazards** 1 **Flammability** 1 **Instability** 0 **Special hazards** -

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

TWA	TWA (time-weighted average)	STEL	STEL (Short Term Exposure Limit)
Ceiling	Maximum limit value	Sk*	Skin designation
+	Sensitizers		

Key literature references and sources for data used to compile the SDS

U.S. Agency for Toxic Substances and Disease Registry (ATSDR)
 U.S. Environmental Protection Agency ChemView Database
 European Food Safety Authority (EFSA)
 U.S. Environmental Protection Agency
 Acute Exposure Guideline Level(s) (AEGL(s))
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act
 U.S. Environmental Protection Agency High Production Volume Chemicals
 Food Research Journal
 Hazardous Substance Database
 International Uniform Chemical Information Database (IUCLID)
 Japan National Institute of Technology and Evaluation (NITE)
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)
 NIOSH (National Institute for Occupational Safety and Health)
 National Library of Medicine's ChemID Plus (NLM CIP)
 National Library of Medicine's PubMed database (NLM PUBMED)
 U.S. National Toxicology Program (NTP)
 New Zealand's Chemical Classification and Information Database (CCID)
 International Organization for Economic Co-operation and Development (OECD) Environment, Health, and Safety Publications
 International Organization for Economic Co-operation and Development (OECD) High Production Volume Chemicals Program
 International Organization for Economic Co-operation and Development (OECD) Screening Information Data Set
 United Nations World Health Organization (WHO)

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Revision Summary

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

End of Safety Data Sheet